

PYTHON FUNCTIONS AND EXPRESSIONS Meta

COMPUTER SCIENCE MENTORS 61A

September 2 – September 5, 2025

Example Timeline

- Teaching tips:

Each problem's "Suggested Time" is not assigned with the 1 hour time for sections in mind. It is more of a note of how long each individual problem may take. That is, if you added all the "Suggested Times" in the coding section, it could very well be greater than 1 hour (especially in more complicated worksheets in the future).

However, the time in brackets will add up to 50 minutes, and is intended to be a suggested outline for your section.

- Intros, Ice Breakers, & Logistics [10 min]

Introduce yourself, and give an overview of what CSM offers. Start off with an icebreaker of your choice!

- Mini-lecture: Python Expressions and Functions [10 min]
- Q1: Potpourri [7 min]

You can skip the question with the if statements, and feel free to skip some parts!

- Q2: Identify Types [7 min]
- Q3: Funky Functions [7 min]
- Q4: Fahrenheit to Celsius [5 min]
- Q5: Score Needed [7 min]
- Q6: WWPD [If Extra Time]

1 Expressions

1. What Would Python Display?

```
>>> 3

3

>>> "cs61a"
```

```
'cs61a'
```

```
>>> x = 3
```

```
>>> x
```

```
3
```

```
>>> x = print("cs61a")
```

```
cs61a
```

```
>>> x
```

```
>>> print(print(print("cs61a")))
```

```
cs61a
```

```
None
```

```
None
```

```
>>> def f1(x):
```

```
...     return x + 1
```

```
>>> f1(3)
```

```
4
```

```
>>> f1(2) + f1(2 + 3)
```

```
9
```

```
>>> def f2(y):
```

```
...     return y / 0
```

```
>>> f2(4)
```

```
ZeroDivisionError: division by zero
```

```
>>> def f3(x, y):
```

```
...     if x > y:
```

```
...         return x
```

```
...     elif x == y:
```

```
...         return x + y
```

```
...     else:
```

```
...         return y
```

```
>>> f3(1, 2)
```

```
2
```

```
>>> f3(5, 5)
```

```
10
```

```
>>> 1 or 2 or 3
```

```
1
```

```
>>> 1 or 0 or 3
```

1

```
>>> 4 and (2 or 1/0)
```

2

```
>>> 0 or (not 1 and 3)
```

False

```
>>> (2 or 1/0) and (False or (True and (0 or 1)))
```

1

Suggested Time: 7 min

- You can probably skip a lot of these as soon as your students get the idea.
- Your students might be new to solving “compound” questions like the last one; Consider giving them tips on how to break down such problems.

2. What would Python display for each expression?

(a) $3 + 4$

7

(b) $'3' + 4$

TypeError

(c) $'3 + 4'$

$'3 + 4'$

(d) $'3' + '4'$

$'34'$

Suggested Time: 7 min, Difficulty: Medium

- Make sure students understand why $'3' + 4$ errors, as well as how string concatenation works in the case of $'3' + '4'$

2 Functions

1. For the following expressions, simplify the operands in the order of evaluation of the entire expression

Example: `add(3, mul(4, 5))`

Order of Evaluation: `add(3, mul(4, 5))` $\rightarrow$ `add(3, 20)` $\rightarrow$ 23

(a) `add(1, mul(2, 3))`

```
add(1, mul(2, 3))
add(1, 6)
7
```

(b) `add(mul(2, 3), add(1, 4))`

```
add(mul(2, 3), add(1, 4))
add(6, add(1, 4))
add(6, 5)
11
```

(c) `max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)`

```
max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)
max(2, add(5, 6), 3, mul(mul(3, 4), 1), 7)
max(2, 11, 3, mul(mul(3, 4), 1), 7)
max(2, 11, 3, mul(12, 1), 7)
max(2, 11, 3, 12, 7)
12
```

Suggested Time: 7 min

- Show students how to tackle problems with nested expressions (like the process in the solution for this problem).
2. Write a function `fahrenheitToCelsius`, which takes in an integer `f` and returns the temperature in Celsius. Hint: the formula for Fahrenheit to Celcius is to subtract 32 and then multiply by 5/9.

```
def fahrenheitToCelsius(f):
    """
    >>> fahrenheitToCelsius(32)
    0.0
    >>> fahrenheitToCelsius(104)
    40.0
    """
```

```
def fahrenheitToCelsius(f):
    return (f - 32) * (5/9)
```

Teaching Tips

1. Talk about the difference between integer and float division. Explain why the outputs of the doctests are decimals instead of integers.

Suggested Time: 5 min, Difficulty: Easy

3. Sally really wants to know what she needs to score to pass her class! Fill out the function `score_needed` that takes in *current_grade*, her current grade, *num_exams*, the number of exams taken so far (not including her final exam), and *pass_grade*, the grade needed to pass. Assume that every exam, including the final exam, has the same weight.

```
def score_needed(current_grade, num_exams, pass_grade):  
    """  
    >>> score_needed(89, 4, 90)  
    94  
    >>> score_needed(65, 2, 70)  
    80  
    >>> score_needed(77, 10, 78)  
    88  
    """  
  
    def score_needed(current_grade, num_exams, pass_grade):  
        total_pts = current_grade * num_exams  
        total_needed = pass_grade * (num_exams + 1)  
        return total_needed - total_pts
```

Suggested Time: 7 min, Difficulty: Medium Make sure that students understand the question prompt and how to come up with the formula.

4. What would Python display?

```
>>> from operator import sub, mul  
>>> def print_sub(x, y):  
...     print('sub')  
...     return sub(x, y)  
>>> def print_mul(x, y):  
...     print('mul')  
...     return mul(x, y)  
>>> print_sub(print_mul(505, 4), 3)  
  
mul  
sub  
2017
```

Suggested Time: 5 min, Difficulty: Medium

- Optional, for if you have time at the end!